

We Perceive What We Expect

1

Our perception of the world around us is not a true depiction of what is actually there. We perceive, to a large extent, what we *expect* to perceive. Our expectations—and therefore our perceptions—are biased by three factors:

- ***the past:*** our experience
- ***the present:*** the current context
- ***the future:*** our goals

PERCEPTION BIASED BY EXPERIENCE

Imagine that you own a large insurance company. You are meeting with a real estate manager, discussing plans for a new campus of company buildings. The campus consists of a row of five buildings, the last two with T-shaped courtyards providing light for the cafeteria and fitness center. If the real estate manager showed you the map shown in [Figure 1.1](#), you would see five black shapes representing the buildings.

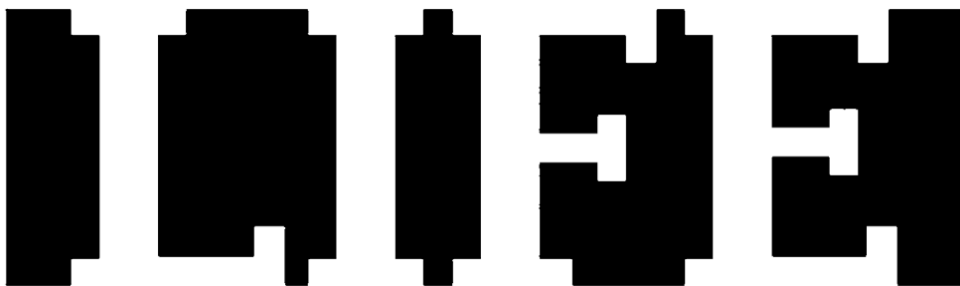


FIGURE 1.1

Building map or word? What you see depends on what you were told to see.

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Now imagine that instead of a real estate manager, you are meeting with an advertising manager. You are discussing a new billboard ad to be placed in certain markets around the country. The advertising manager shows you the same image, but in this scenario the image is a sketch of the ad, consisting of a single word. In this scenario, you see a word, clearly and unambiguously.

When your perceptual system has been primed to see building shapes, you see building shapes, and the white areas between the buildings barely register in your perception. When your perceptual system has been primed to see text, you see text, and the black areas between the letters barely register.

A relatively famous example of how priming the mind can affect perception is a sketch, supposedly by R. C. James,¹ that initially looks to most people like a random splattering of ink (see Fig. 1.2). Before reading further, look at the sketch.



FIGURE 1.2

Image showing the effect of mental priming of the visual system. What do you see?

Only after you are told that it is a Dalmatian dog sniffing the ground near a tree can your visual system organize the image into a coherent picture. Moreover, once you've "seen" the dog, it is hard to go back to seeing the image as a random collection of spots.

¹ Published in Marr D. (1982) *Vision*. W. H. Freeman, New York, NY, p. 101, Figure 3-1.

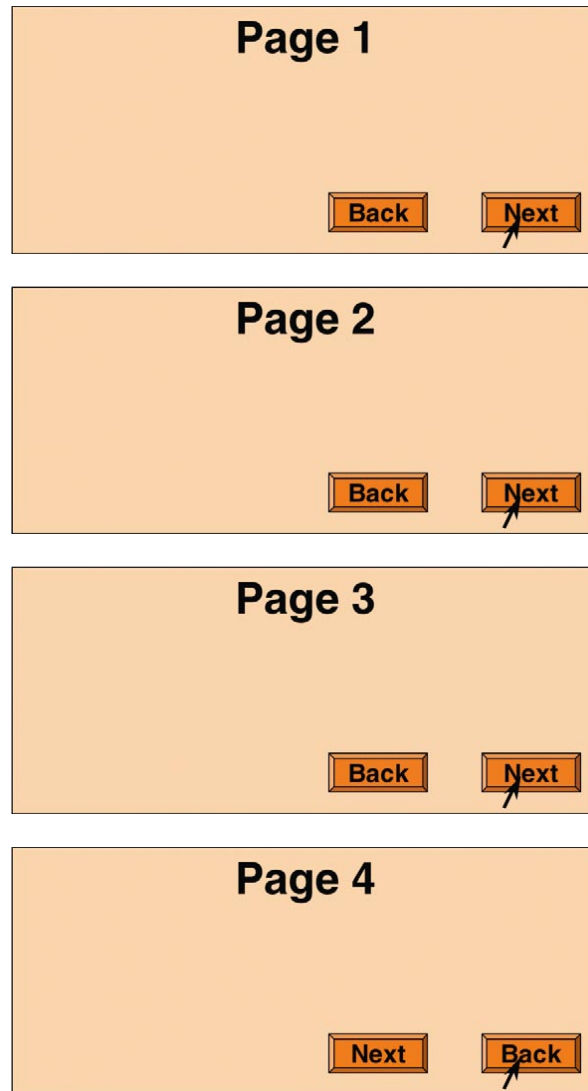


FIGURE 1.3

The “Next” button is perceived to be in a consistent location, even when it isn’t.

The examples above are visual. Experience can also bias other types of perception, such as sentence comprehension. For example, the headline “New Vaccine Contains Rabies” would probably be understood differently by people who had recently heard stories about contaminated vaccines than by people who had recently heard stories about successful uses of vaccines to fight diseases.

Users of computer software and Web sites often click buttons or links without looking carefully at them. Their perception of the display is based more on what

their past experience leads them to expect than on what is actually on the screen. This sometimes confounds software designers, who expect users to see what is on the screen. But that isn't how perception works.

For example, if the positions of the “Next” and “Back” buttons on the last page of a multipage dialog box² switched, many people would not immediately notice the switch (see Fig. 1.3). Their visual system would have been lulled into inattention by the consistent placement of the buttons on the prior several pages. Even after unintentionally going backward a few times, they might continue to perceive the buttons in their standard locations. This is why “place controls consistently” is a common user interface design guideline.

Similarly, if we are trying to find something, but it is in a different place or looks different from usual, we might miss it even though it is in plain view because experience tunes us to look for expected features in expected locations. For example, if the “Submit” button on one form in a Web site is shaped differently or is a different color from those on other forms on the site, users might not find it. This expectation-induced blindness is discussed further later in this chapter, in the section on how our *goals* affect perception.

PERCEPTION BIASED BY CURRENT CONTEXT

When we try to understand how our visual perception works, it is tempting to think of it as a bottom-up process, combining basic features such as edges, lines, angles, curves, and patterns into figures and ultimately into meaningful objects. To take reading as an example, you might assume that our visual system first recognizes shapes as letter and then combines letters into words, words into sentences, and so on.

But visual perception—reading in particular—is not strictly a bottom-up process. It includes top-down influences too. For example, the word in which a character appears may affect how we identify the character (see Fig. 1.4).

Similarly, our overall comprehension of a sentence or of a paragraph can even influence what words we see in it. For example, the same letter sequence can be



FIGURE 1.4

The same character is perceived as H or A depending on the surrounding letters.

²Multi step dialog boxes are called “wizards” in user interface jargon.

Fold napkins. *Polish silverware.* Wash dishes.

French napkins. *Polish silverware.* German dishes.

FIGURE 1.5

The same phrase is perceived differently depending on the list it appears in.

read as different words depending on the meaning of the surrounding paragraph (see Fig. 1.5).

This biasing of perception by the surrounding context works *between* different senses too. Perceptions in any of our five senses may affect simultaneous perceptions in any of our other senses. For example:

- What we see can be biased by what we are hearing, and *vice versa*
- What we feel with our tactile sense can be biased by what we are hearing, seeing, or smelling

Later chapters explain how visual perception, reading, and recognition function in the human brain. For now, I will simply say that the pattern of neural activity that corresponds to recognizing a letter, a word, a face, or any object includes input from neural activity stimulated by the context. This context includes other nearby perceived objects and events, and even reactivated memories of previously perceived objects and events.

Context biases perception not only in people but also in lower animals. A friend of mine often brought her dog with her in her car when running errands. One day, as she drove into her driveway, a cat was in the front yard. The dog saw it and began barking. My friend opened the car door and the dog jumped out and ran after the cat, which turned and jumped through a bush to escape. The dog dove into the bush but missed the cat. The dog remained agitated for some time afterward.

Thereafter, for as long as my friend lived in that house, whenever she arrived at home with her dog in the car, he would get excited, bark, jump out of the car as soon as the door was opened, dash across the yard, and leap into the bush. There was no cat, but that didn't matter. Returning home in the car was enough to make the dog see one—perhaps even smell one. However, walking home, as the dog did after being taken for his daily walk, did not evoke the “cat mirage.”

PERCEPTION BIASED BY GOALS

In addition to being biased by our *past* experience and the *present* context, our perception is influenced by our goals and plans for the *future*. Specifically, our goals filter our perceptions: things unrelated to our goals tend to be filtered out preconsciously, never registering in our conscious minds.

For example, when people navigate through software or a Web site, seeking information or a specific function, they don't read carefully. They scan screens quickly

and superficially for items that seem related to their goal. They don't simply *ignore* items unrelated to their goals; they often don't even *notice* them.

To see this, flip briefly to the next page and look in the toolbox (Fig. 1.6) for *scissors*, and then immediately flip back to this page. Try it now.

Did you spot the scissors? Now, without looking back at the toolbox, can you say whether there is a screwdriver in the toolbox too?

Our goals filter our perceptions in other perceptual senses as well as in vision. A familiar example is the “cocktail party” effect. If you are conversing with someone at a crowded party, you can focus your attention to hear mainly what he or she is saying even though many other people are talking near you. The more interested you are in the conversation, the more strongly your brain filters out surrounding chatter. If you are bored by what your conversational partner is saying, you will probably hear much more of the conversations around you.

The effect was first documented in studies of air-traffic controllers, who were able to carry on a conversation with the pilots of their assigned aircraft even though many different conversations were occurring simultaneously on the same radio frequency, coming out of the same speaker in the control room (Arons, 1992). Research suggests that our ability to focus on one conversation among several simultaneous ones depends not only on our interest level in the conversation but also on objective factors such as the similarity of voices in the cacophony, the amount of general “noise” (e.g., clattering dishes or loud music), and the predictability of what your conversational partner is saying (Arons, 1992).

This filtering of perception by our goals is particularly true for adults, who tend to be more focused on goals than children are. Children are more stimulus driven: their perception is less filtered by their goals. This characteristic makes them more distractible than adults, but it also makes them less biased as observers.

A parlor game demonstrates this age difference in perceptual filtering. It is similar to the “look in the toolbox” exercise. Most households have a catch-all drawer for kitchen implements or tools. From your living room, send a visitor to the room where the catch-all drawer is, with instructions to fetch you a specific tool, such as measuring spoons or a pipe wrench. When the person returns with the tool, ask whether another specific tool was in the drawer. Most adults will not know what else was in the drawer. Children—if they can complete the task without being distracted by all the cool stuff in the drawer—will often be able to tell you more about what else was there.

Perceptual filtering can also be seen in how people navigate Web sites. Suppose I put you on the home page of New Zealand's University of Canterbury (see Fig. 1.7) and asked you to print out a map of the campus showing the computer science department. You would scan the page and probably quickly click one of the links that share words with the goal that I gave you: *Departments* (top left), *Departments and Colleges* (middle left), or *Campus Maps* (bottom right). If you're a “search” person, you might instead go right to the Search box (middle right), type words related to the goal, and click “Go.”



FIGURE 1.6

Toolbox: Are there scissors here?

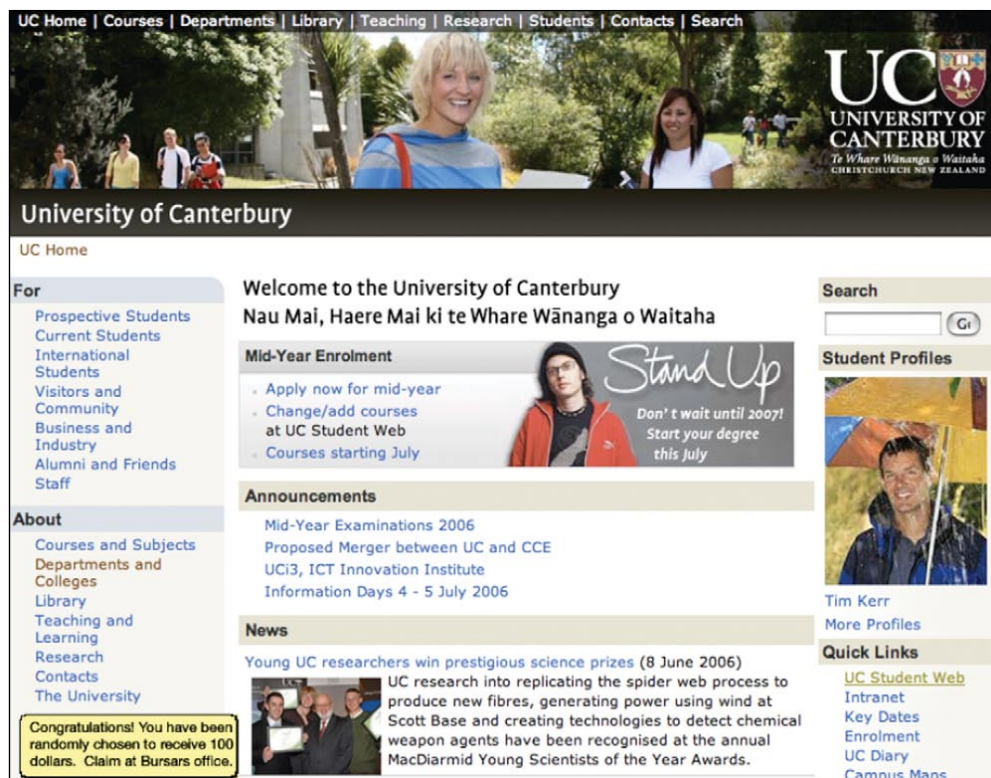


FIGURE 1.7

University of Canterbury home page: Navigating Web sites includes perceptual filtering.

Whether you browse or search, it is likely that you would leave the home page without noticing that you were randomly chosen to win \$100 (bottom left). Why? Because that was not related to your *goal*.

What is the mechanism by which our current goals bias our perception? There are two:

- ***Influencing where we look.*** Perception is active, not passive. We constantly move our eyes, ears, hands, and so on, so as to sample exactly the things in our environment that are most relevant to what we are doing or about to do (Ware, 2008). If we are looking on a Web site for a campus map, our eyes and pointer-controlling hand are attracted to anything that might lead us to that goal. We more or less ignore anything unrelated to our goal.
- ***Sensitizing our perceptual system to certain features.*** When we are looking for something, our brain can prime our perception to be especially sensitive to features of what we are looking for (Ware, 2008). For example, when we are looking for a red car in a large parking lot, red cars will seem to pop out as we scan the lot, and cars of other colors will barely register in our consciousness, even though we do in some sense “see” them. Similarly, when we are trying to find our spouse in a dark, crowded room, our brain “programs” our auditory system to be especially sensitive to the combination of frequencies that make up his or her voice.

DESIGN IMPLICATIONS

All these sources of perceptual bias of course have implications for user interface design. Here are three.

Avoid ambiguity

Avoid ambiguous information displays, and test your design to verify that all users interpret the display in the same way. Where ambiguity is unavoidable, either rely on standards or conventions to resolve it, or prime users to resolve the ambiguity in the intended way.

For example, computer displays often shade buttons and text fields to make them look raised in relation to the background surface (see [Fig. 1.8](#)). This appearance



FIGURE 1.8

Buttons on computer screens are often shaded to make them look three dimensional, but the convention only works if the simulated light source is assumed to be on the upper left.

relies on a convention, familiar to most experienced computer users, that the light source is at the top left of the screen. If an object were depicted as lit by a light source in a different location, users would not see the object as raised.

Be consistent

Place information and controls in consistent locations. Controls and data displays that serve the same function on different pages should be placed in the same position on each page on which they appear. They should also have the same color, text fonts, shading, and so on. This consistency allows users to spot and recognize them quickly.

Understand the goals

Users come to a system with goals they want to achieve. Designers should understand those goals. Realize that users' goals may vary, and that their goals strongly influence what they perceive. Ensure that at every point in an interaction, the information users need is available, prominent, and maps clearly to a possible user goal, so users will notice and use the information.

Our Vision is Optimized to See Structure

2

Early in the twentieth century, a group of German psychologists sought to explain how human visual perception works. They observed and catalogued many important visual phenomena. One of their basic findings was that human vision is holistic: Our visual system automatically imposes structure on visual input and is wired to perceive whole shapes, figures, and objects rather than disconnected edges, lines, and areas. The German word for “shape” or “figure” is *Gestalt*, so these theories became known as the Gestalt principles of visual perception.

Today’s perceptual and cognitive psychologists regard the Gestalt theory of perception as more of a *descriptive* framework than an *explanatory* and *predictive* theory. Today’s theories of visual perception tend to be based heavily on the neurophysiology of the eyes, optic nerve, and brain (see Chapters 4–7).

Not surprisingly, the findings of neurophysiological researchers support the observations of the Gestalt psychologists. We really are—along with other animals—“wired” to perceive our surroundings in terms of whole objects (Stafford & Webb, 2005; Ware, 2008). Consequently, the Gestalt principles are still valid—if not as a fundamental explanation of visual perception, at least as a framework for describing it. They also provide a useful basis for guidelines for graphic and user interface design (Soegaard, 2007).

For present purposes, the most important Gestalt principles are: Proximity, Similarity, Continuity, Closure, Symmetry, Figure/Ground, and Common Fate. In the following sections, I describe each principle and provide examples from both static graphic design and user interface design.

GESTALT PRINCIPLE: PROXIMITY

The principle of *Proximity* is that the relative distance between objects in a display affects our perception of whether and how the objects are organized into subgroups. Objects that are near each other (relative to other objects) appear grouped, while those that are farther apart do not.

In Figure 2.1, the stars on the left are closer together horizontally than they are vertically, so we see three rows of stars, while the stars on the right are closer together vertically than they are horizontally, so we see three columns.

The Proximity principle has obvious relevance to the layout of control panels or data-forms in software, Web sites, and electronic appliances. Designers often separate groups of on-screen controls and data-displays by enclosing them in group boxes or by placing separator lines between groups (see Fig. 2.2).

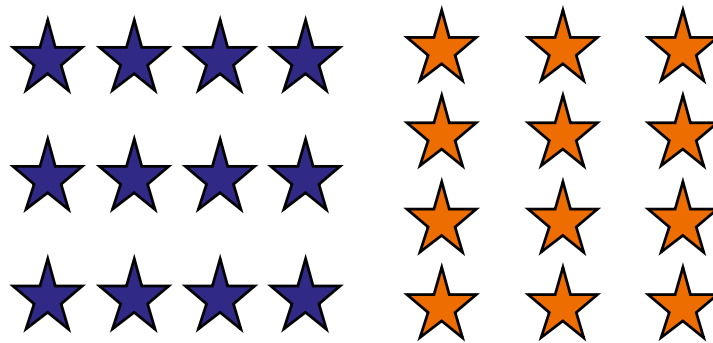


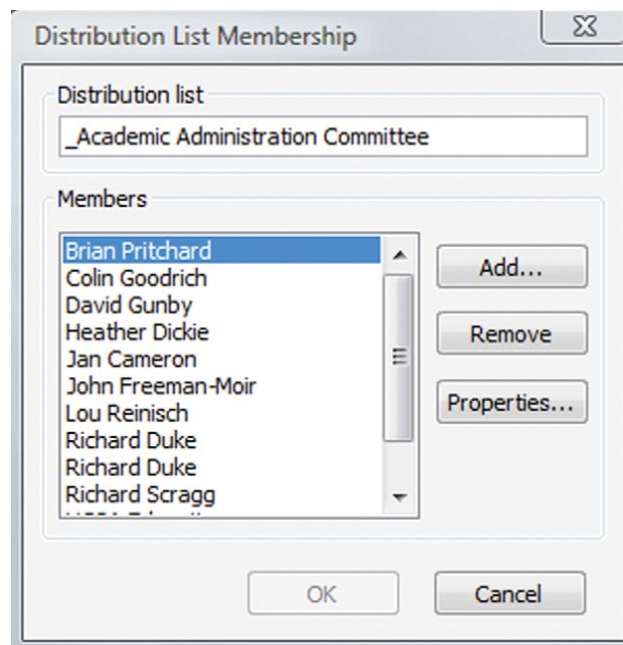
FIGURE 2.1

Proximity: Items that are closer appear grouped. Left: rows, Right: columns.



FIGURE 2.2

In Outlook's Distribution List Membership dialog box, list buttons are in a group box, separate from the window-control buttons.



However, according to the Proximity principle, items on a display can be visually grouped simply by spacing them closer together to each other than to other controls, without group boxes or visible borders (see Fig. 2.3). Many graphic design experts recommend this approach in order to reduce visual clutter and code size in a user interface (Mullet & Sano, 1994).

Conversely, if controls are *poorly* spaced, e.g., if connected controls are too far apart, people will have trouble perceiving them as related, making the software harder to learn and remember. For example, the Discreet Software Installer displays six horizontal pairs of radiobuttons, each representing a two-way choice, but their spacing, due to the Proximity principle, makes them appear to be two vertical sets of radiobuttons, each representing a six-way choice, at least until users try them and learn how they operate (see Fig. 2.4).

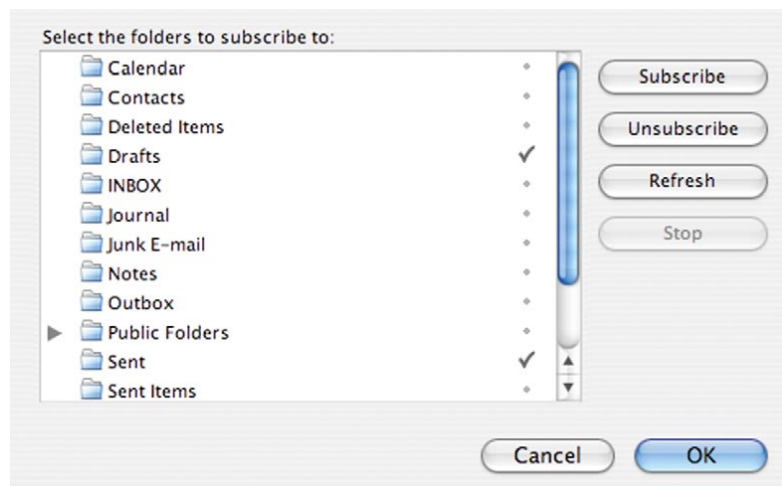


FIGURE 2.3

In Mozilla Thunderbird's Subscribe Folders dialog box, controls are grouped using the Proximity principle.



FIGURE 2.4

In Discreet's Software Installer, poorly spaced radiobuttons look grouped in vertical columns.

GESTALT PRINCIPLE: SIMILARITY

Another factor that affects our perception of grouping is expressed in the Gestalt principle of *Similarity*: Objects that look similar appear grouped, all other things being equal. In [Figure 2.5](#), the slightly larger, “hollow” stars are perceived as a group.

The Page Setup dialog box in Mac OS applications uses the Similarity and Proximity principles to convey groupings (see [Fig. 2.6](#)). The three very similar and tightly spaced

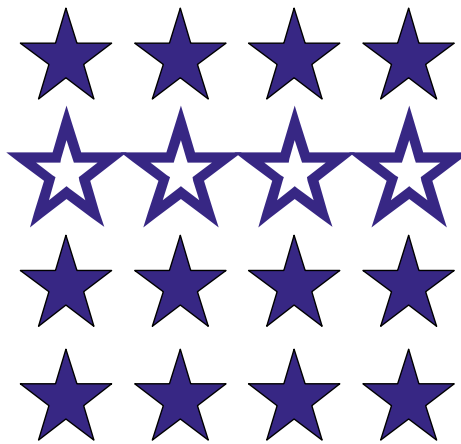


FIGURE 2.5

Similarity: Items appear grouped if they look more similar to each other than to other objects.

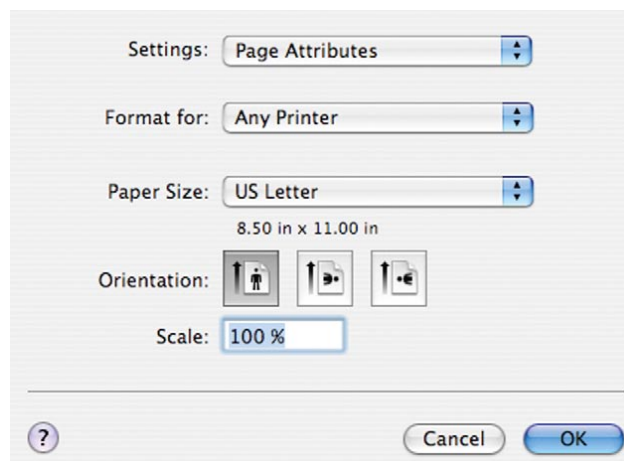


FIGURE 2.6

Mac OS Page Setup dialog box: The Similarity and Proximity principles are used to group the Orientation settings.

Orientation settings are clearly intended to appear grouped. The three menus are not so tightly spaced but look similar enough that they appear related even though that probably wasn't intended.

Similarly, the text fields in a form at the Web site of book publisher Elsevier are organized into an upper group of seven (with three subgroups) for the address, a group of three split fields for phone numbers, and two single text fields. The four menus, in addition to being data fields, help separate the text field groups (see Fig. 2.7). By contrast, the labels are too far from their fields to seem connected to them.

The screenshot shows a web form with the following elements:

- Title (Mr, Ms, Dr etc): A dropdown menu with the text "**Please Select**".
- First name: A single-line text input field.
- Last name: A single-line text input field.
- Job title: A single-line text input field.
- Institution/Organisation: A single-line text input field.
- Number and Street: A single-line text input field.
- City: A single-line text input field.
- State/County: A single-line text input field.
- Zip Code/Postal Code: A single-line text input field.
- Country: A dropdown menu with the text "**Please Select**".
- Work phone: Two side-by-side single-line text input fields.
- Home phone: Two side-by-side single-line text input fields.
- Fax: Two side-by-side single-line text input fields.
- How did you find out about this Web site: A dropdown menu with the text "Please select".
- Other: A single-line text input field.
- Please select the option which most closely describes you as a customer: A dropdown menu with the text "Please select".
- E-mail: A single-line text input field.

Red arrows on the right side of the form point to the dropdown menus and the split phone number fields, highlighting their visual grouping.

FIGURE 2.7

Online form at Elsevier.com: Similarity makes the text fields appear grouped.

GESTALT PRINCIPLE: CONTINUITY

In addition to the two Gestalt principles concerning our tendency to organize objects into groups, several Gestalt principles describe our visual system's tendency to resolve ambiguity or fill in missing data in such a way as to perceive whole objects. The first such principle, the principle of *Continuity*, states that our visual perception is biased to perceive continuous forms rather than disconnected segments.

For example, on the left side of [Figure 2.8](#), we automatically see two crossing lines—one blue and one orange. We don’t see two separate orange segments and two separate blue ones, and we don’t see a blue-and-orange V on top of an upside-down orange-and-blue V. On the right side of [Figure 2.8](#), we see a sea monster in water, not three pieces of one.

A well-known example of the use of the Continuity principle in graphic design is the IBM® logo. It consists of disconnected blue patches, and yet it is not at all ambiguous; it is easily seen as three bold letters, perhaps viewed through something like venetian blinds (see [Fig. 2.9](#)).

Slider controls are a user-interface example of the Continuity principle. We see a slider as depicting a single range controlled by a handle that appears somewhere on the slider, not as two separate ranges separated by the handle (see [Fig. 2.10A](#)). Even displaying different colors on each side of a slider’s handle doesn’t completely “break” our perception of a slider as one continuous object, although ComponentOne’s choice of strongly contrasting colors (gray vs. red) certainly strains that perception a bit (see [Fig. 2.10B](#)).

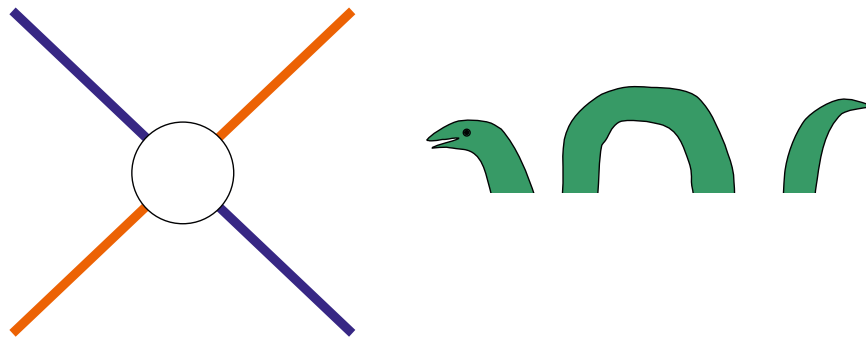


FIGURE 2.8

Continuity: Human vision is biased to see continuous forms, even adding missing data if necessary.



FIGURE 2.9

The IBM company logo uses the Continuity principle to form letters from disconnected patches.

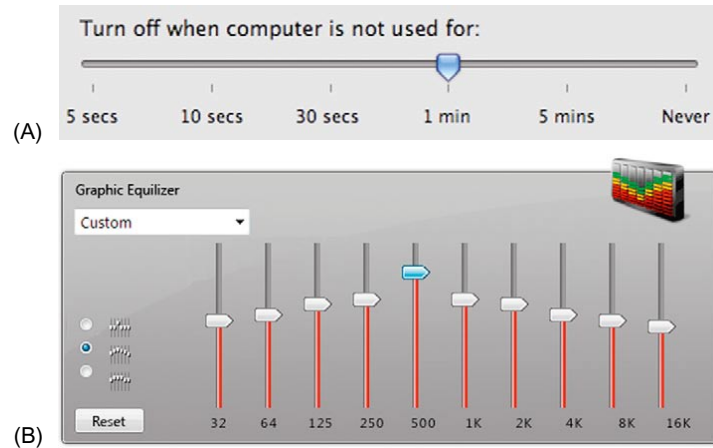


FIGURE 2.10

Continuity: We see a slider as a single slot with a handle somewhere on it, not as two slots separated by a handle. (A) Mac OS, (B) ComponentOne.

GESTALT PRINCIPLE: CLOSURE

Related to Continuity is the Gestalt principle of *Closure*, which states that our visual system automatically tries to close open figures so that they are perceived as whole objects rather than separate pieces. Thus, we perceive the disconnected arcs on the left of Figure 2.11 as a circle.

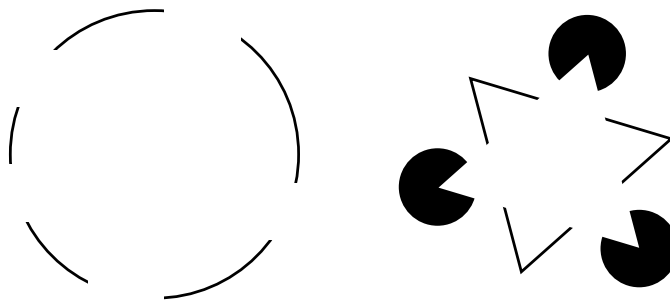


FIGURE 2.11

Closure: Human vision is biased to see whole objects, even when they are incomplete.

Our visual system is so strongly biased to see objects that it can even interpret a totally blank area as an object. We see the combination of shapes on the right of Figure 2.11 as a white triangle overlapping another triangle and three black circles, even though the figure really only contains three V-shapes and three black pac-men.

The Closure principle is often applied in graphical user interfaces (GUIs). For example, GUIs often represent collections of objects—e.g., documents or messages—as *stacks* (see Fig. 2.12). Just showing one whole object and the edges of others “behind” it is enough to make users perceive a stack of objects, all whole.



FIGURE 2.12

Icons depicting stacks of objects exhibit the Closure principle: partially visible objects are perceived as whole.

GESTALT PRINCIPLE: SYMMETRY

A third fact about our tendency to see objects is captured in the Gestalt principle of *Symmetry*. It states that we tend to parse complex scenes in a way that reduces the complexity. The data in our visual field usually has more than one possible interpretation, but our vision automatically organizes and interprets the data so as to simplify it and give it symmetry.

For example, we see the complex shape on the left of Figure 2.13 as two overlapping diamonds, not as two touching corner bricks or a pinch-waist octahedron with a square in its center. A pair of overlapping diamonds is simpler than the other two interpretations shown on the right of Figure 2.13: it has fewer sides and more symmetry than the other two interpretations.

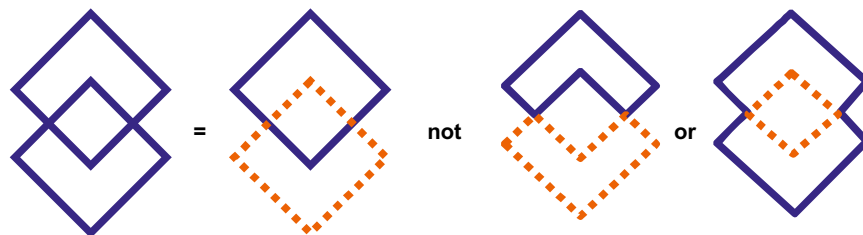


FIGURE 2.13

Symmetry: The human visual system tries to resolve complex scenes into combinations of simple, symmetrical shapes.

In printed graphics and on computer screens, our visual system’s reliance on the symmetry principle can be exploited to represent three dimensional objects on a two dimensional display. This can be seen in a cover illustration for Paul Thagard’s book *Coherence in Thought and Action* (Thagard, 2002; see Fig. 2.14) and in three-dimensional depiction of a cityscape (see Fig. 2.15).



FIGURE 2.14

The cover of the book *Coherence in Thought and Action* (Thagard, 2002) uses the Symmetry, Closure, and Continuity principles to depict a cube.



FIGURE 2.15

Symmetry: The human visual system parses very complex two dimensional images into three dimensional scenes.

GESTALT PRINCIPLE: FIGURE/GROUND

The next Gestalt principle that describes how our visual system structures the data it receives is *Figure/Ground*. This principle states that our mind separates the visual field into the figure (the foreground) and ground (the background). The foreground

consists of those elements of a scene that are the object of our primary attention, and the background is everything else.

The Figure/Ground principle also specifies that the visual system's parsing of scenes into figure and ground is influenced by characteristics of the scene. For example, when a small object or color patch overlaps a larger one, we tend to perceive the smaller object as the figure and the larger object as the ground (see [Fig. 2.16](#)).

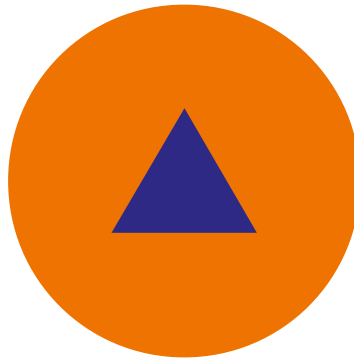


FIGURE 2.16

Figure/Ground: When objects overlap, we see the smaller as figure on ground.

However, our perception of figure vs. ground is not completely determined by scene characteristics. It also depends on the viewer's focus of attention. Dutch artist M. C. Escher exploited this phenomenon to produce ambiguous images in which figure and ground switch roles as our attention shifts (see [Fig. 2.17](#)).



FIGURE 2.17

M. C. Escher exploited figure/ground ambiguity in his art.

In user interface and Web design, the Figure/Ground principle is often used to place an impression-inducing background “behind” the primary displayed content (see Fig. 2.18). The background can convey information—e.g., the user’s current location—or it can suggest a theme, brand, or mood for interpretation of the content.



FIGURE 2.18

Figure/Ground is used at AndePhotos.com to display a thematic watermark “behind” content.

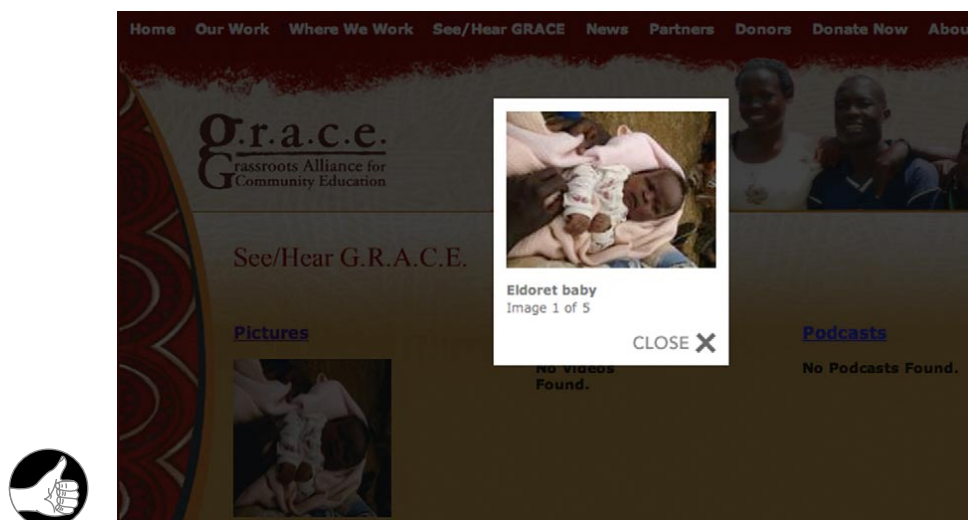


FIGURE 2.19

Figure/Ground is used at GRACEUSA.org to pop up a photo “over” the page content.

Figure/Ground is also often used to pop up information over other content. Content that was formerly the figure—the focus of the users’ attention—temporarily becomes the *background* for new information, which appears briefly as the new *figure* (see Fig. 2.19). This approach is usually better than temporarily *replacing* the old information with the new information, because it provides context that helps keep people oriented regarding their place in the interaction.

GESTALT PRINCIPLES: COMMON FATE

The previous six Gestalt principles concerned perception of static (un-moving) figures and objects. One final Gestalt principle—Common Fate—concerns moving objects. The Common Fate principle is related to the Proximity and Similarity principles: Like them it affects whether we perceive objects as grouped. The Common Fate principle states that objects that move together are perceived as grouped or related.

For example, in a display showing dozens of pentagons, if seven of them wiggled back and forth in synchrony, people would see them as a related group, even if the wiggling pentagons were separated from each other and looked no different from all the other pentagons (see Fig. 2.20).

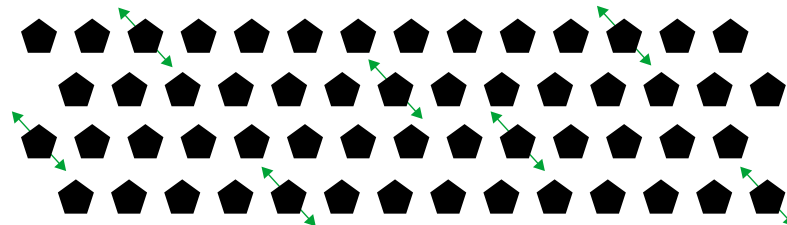


FIGURE 2.20

Common Fate: Items appear grouped or related if they move together.

Common motion—implying common fates—is used in some animations to show relationships between entities. For example, GapMinder graphs animate dots representing nations to show changes over time in various factors of economic development. Countries that move together share development histories (see Fig. 2.21).

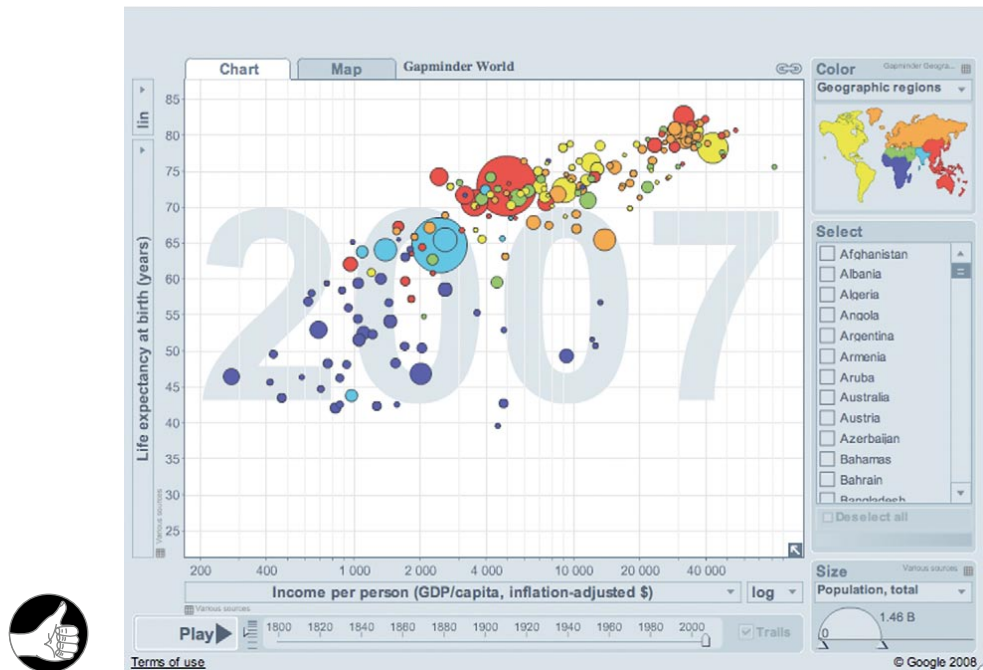


FIGURE 2.21

Common fate: GapMinder animates dots to show which nations have similar development histories.

GESTALT PRINCIPLES: COMBINED

Of course, in real-world visual scenes, the Gestalt principles work in concert, not in isolation. For example, a typical Mac OS desktop usually exemplifies six of the seven principles described above (excluding Common Fate): Proximity, Similarity, Continuity, Closure, Symmetry, and Figure/Ground (see Fig. 2.22). On a typical desktop, Common Fate is used (along with similarity) when a user selects several files or folders and drags them as a group to a new location (see Fig. 2.23).

With all these Gestalt principles operating at once, *unintended* visual relationships can be implied by a design. A recommended practice, after designing a display, is to view it with each of the Gestalt principles in mind—Proximity, Similarity, Continuity, Closure, Symmetry, Figure/Ground, and Common Fate—to see if the design suggests any relationships between elements that you do *not* intend.

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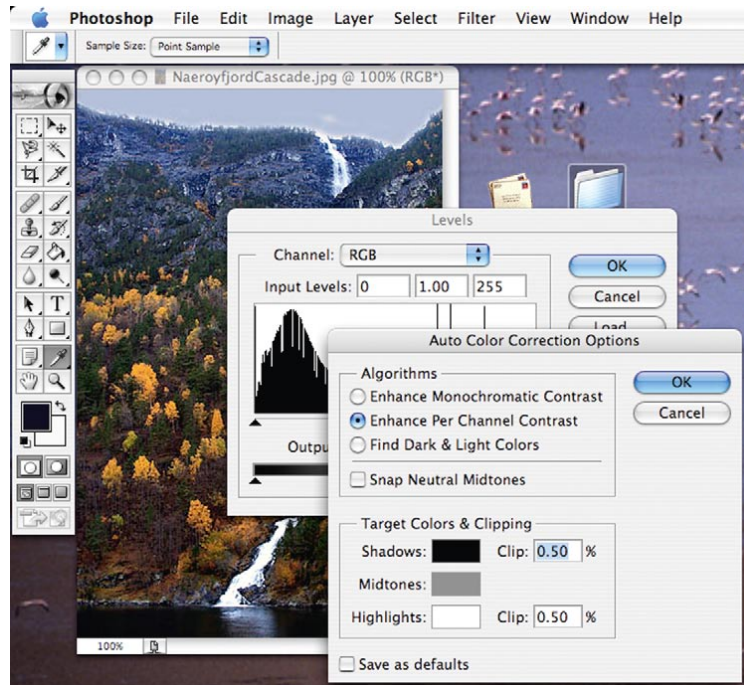


FIGURE 2.22

All of the Gestalt principles except Common Fate play a role in this portion of a Mac OS desktop.

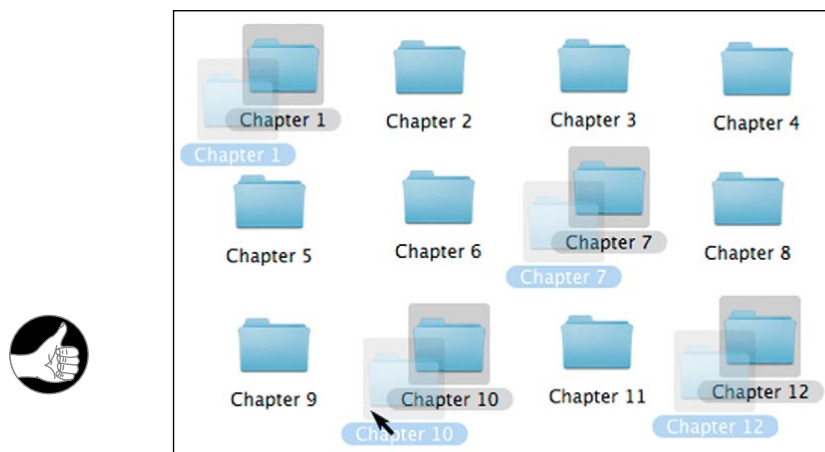


FIGURE 2.23

Similarity and Common Fate: When users drag folders that they have selected, common highlighting and motion make the selected folders appear grouped.